Exp No: 12

EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

OBJECTIVES

- To write an Embedded C program to multiply two 16-bit binary numbers using Keil software.
- To obtain and store the resulting 32-bit product.
- To display the 32-bit result through four 8-bit ports (P0, P1, P2, and P3).

PROGRAM

```
#include <reg51.h> void main()

{

while (1)

{

unsigned int num1, num2; unsigned long int product; num1 = 0x2222;

num2 = 0xBBBB;

product = (unsigned long int)num1 * num2; P0= product &0xff;

P1= (product & 0xff00)>>8;

P2 =(product & 0xff0000)>>16; P3 =(product & 0xff000000)>>24;

}

}
```

OUTPUT